

Shanmukha Bhargav Bhamidipati

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SUMMARY

Software & Data Engineer with 2+ years of experience building cloud data pipelines, Snowflake data warehouse solutions, backend APIs, and enterprise applications across nonprofit and enterprise environments. Strong in SQL, Python, Snowflake, AWS, and Azure, with hands-on experience in data modeling, workflow automation, reporting systems, and production support. Proven ability to translate business requirements into scalable technical solutions for analytics, operations, and client-facing use cases.

SKILLS

- **Programming & Data Processing:** Python, SQL, PySpark, Java, C#, JavaScript.
- **Big Data & Data Engineering:** PySpark (DataFrames, Spark SQL, Transformations, Joins, Aggregations), Apache Spark, Databricks, Delta Lake, Parquet, ETL / ELT Pipelines, Data Modeling, Data Warehousing, Batch Processing, Incremental Processing, ETL Testing, Data Validation, Data Quality Checks, Schema Validation & Enforcement, Data Reconciliation, Referential Integrity, Business Rule Validation, Pipeline Monitoring, DBT.
- **Cloud & Data Platforms:** AWS (S3, Lambda, EC2), Azure (ADF, ADLS Gen2), Snowflake
- **Tools & Orchestration:** Databricks Workflows, Azure Data Factory, Docker, Jenkins, Git/GitHub
- **Backend & APIs:** Springboot, .NET Core, Flask, REST APIs, Microservices
- **Libraries:** Pandas, Numpy, Scikit-learn, Pydantic

EXPERIENCE

Data Engineer - H.U.G Reading Program

February 2025 – June 2026

- Aggregated teacher-logged data in **SQL** from each student's daily one-hour reading session to calculate weekly lesson-completion rates for parent progress reports
- Built word-recognition accuracy logic in **SQL** by comparing student responses to the same word across multiple session presentations to measure retention over time
- Calculated week-over-week learning-pace metrics in **SQL** by comparing each student's current-week performance against their prior-week session data
- Helped execute a **MySQL-to-Snowflake** migration for student and teacher session data collected across partner schools, simplifying long-term data management
- Performed post-migration **validation** in **Snowflake**, confirming migrated student and teacher records matched the original **MySQL** source data before reports relied on it
- Built internal **Power BI** reports tracking active-teacher count, partner-school enrollment, and total student enrollment across the program for organizational visibility
- Developed an **HTML** and **JavaScript** web view to present internal **Power BI** metrics in a simplified, non-technical format for program staff and stakeholders
- Standardized session-data logging practices across teacher teams in **Snowflake** spanning multiple partner schools to improve consistency for downstream reports

Associate Software Engineer - Hexaware Technologies

February 2022 – September 2023

- Developed backend services and data processing components using **Springboot** and **SQL** for enterprise applications.
- Optimized complex **SQL queries** and **improved data processing performance** for large-scale transactional systems.

PROJECTS

Portfolio Agent | Python, Flask, OpenAI API, Pydantic

- Built an AI-powered Flask application that converts resume content and portfolio requirements into a standalone personal portfolio website.
- Implemented structured resume parsing, JSON schema validation, and automatic retry logic to improve reliability and consistency of generated outputs.
- Added HTML validation and content-generation safeguards to ensure cleaner final website structure and more dependable end-to-end generation results.

Ravenstack Analytics Platform | Data Engineer (dbt / Databricks)

- Designed an end-to-end SaaS analytics pipeline using dbt Core on Databricks with Bronze-Silver-Gold architecture for subscription, churn, usage, and support-ticket datasets.
- Built modular transformation models and analytics-ready datasets to improve metric consistency and support scalable reporting across business workflows.
- Applied modern data engineering practices to structure raw data into curated layers that enabled clearer downstream analysis and performance tracking.

NYC Taxi Data Analytics Platform | Data Engineer (Azure)

- Developed modular Azure data pipelines with automated validation workflows to support scalable analytics on high-volume transportation datasets.
- Designed processing logic for incremental updates and data quality checks, improving maintainability and consistency across the pipeline.
- Structured the platform to support reliable ingestion, transformation, and reporting workflows for large-scale analytical use cases.

Real-Time Dubbing Enhancement with Wav2Lip – Machine Learning Engineer | Team of 3

- Built a real-time backend pipeline in a team of 3 to improve audio-video synchronization performance for dubbing enhancement workflows.
- Improved synchronization accuracy by 35% while supporting smoother coordination between backend processing and frontend integration components.
- Containerized the workflow using Docker to simplify development, improve portability, and support more consistent testing across environments.

EDUCATION

Florida Atlantic University– Boca Raton, FL

May 2025

Master's in Computer Science, Minor in Artificial Intelligence, GPA: 3.86

- Relevant coursework: Introduction to Data Science, Software Engineering, Analysis of Algorithms, Natural Language Processing, Information Retrieval, Artificial Intelligence, Data Mining and Machine Learning, CodePath Intermediate Data Structures, Cloud Computing, Cloud Security.

CERTIFICATIONS

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| • Microsoft Azure Fundamentals, AZ-900 | February 2023 |
| • Amazon AWS Cloud Practitioner | October 2025 |
| • Amazon AWS AI Practitioner | October 2025 |